

# Dimitris Mouris

West Los Angeles, CA 90025, USA

☎ (+1) (302) 407-2887 | ✉ jimouris@udel.edu | 🏠 jimouris.com | 🌐 jimouris | 📺 jimouris | 🐦 @jimouris | 📧 FaHg3QoAAAAJ

*Privacy engineer building systems that protect people at scale, motivated by the impact of cryptography in society and on humanity.*

## Research & Work Experience

### Snap Inc.

Santa Monica, CA, USA

PRIVACY ENGINEER

March 2026 – Now

Design & implement privacy-preserving systems and AI tooling at scale, including leading Snap's privacy knowledge base, powering automated AI privacy reviews, privacy incident response, cross-team AI assistants, and AI validation and code-scanning tools. Lead the privacy strategy for internal agentic AI tools, establishing policy and framework for responsible data use and minimization. Partner with Snap's Growth, Lenses, Specs, Safety, Snapchat+, and other teams to embed *Privacy by Design* from scoping through launch, protecting millions of users.

### Nillion (Nilogy Inc.)

Remote

HEAD OF CRYPTOGRAPHY

June 2025 – March 2026

SENIOR CRYPTOGRAPHY RESEARCHER (R&D)

January 2024 – June 2025

Led cryptography R&D, advancing zkTLS, privacy-preserving LLM protocols with MPC and FHE, and large-scale aggregation primitives (e.g., Mastic, PLASMA). Supported the MPC-based database and TEE-based Confidential VM, alongside papers, talks, demos, and partnerships.

🔒 zkTLS 1, 2, 3   🔒 MPC-TEEs,   🔒 Wave,   🔒 Ripple,   🔒 Curl,   🔒 FHE Summit

### University of Delaware

Newark, DE, USA

RESEARCH & TEACHING ASSISTANT

February 2019 – December 2023

Applied Cryptography • Privacy • Secure Multiparty Computation • Homomorphic Encryption • Zero-knowledge Proofs

### Meta Inc.

Bellevue, WA, USA & Remote

PART-TIME STUDENT RESEARCHER (REMOTE)

September 2022 – December 2022

RESEARCH ENGINEER INTERN

June 2022 – August 2022

Research and implementation of *Delegated Private Matching*, enabling attribution and personalization.   📖 Blog,   🌐 Private-ID

### Amazon Web Services (AWS)

Palo Alto, CA, USA

SOFTWARE DEVELOPMENT ENGINEER (SDE) INTERN

June 2021 – August 2021

Development of Role-Based Access Control (RBAC) for Amazon Redshift to manage permissions and security privileges.

### Athena Research & Innovation Center

Athens, Greece

RESEARCH ASSISTANT & SOFTWARE ENGINEER

September 2017 – December 2018

*My Health My Data (MHMD)* project: Design and implementation of an end-to-end framework for privacy-preserving medical data analytics (i.e., histograms and decision trees) using secure multi-party computation (MPC).

🌐 smpc-analytics

## Education

### Department of Electrical & Computer Engineering, University of Delaware

Newark, DE, USA

DOCTOR OF PHILOSOPHY (PHD) IN ELECTRICAL AND COMPUTER ENGINEERING

February 2019 – December 2023

- Research Topic: *Private and Verifiable Computation*
- Privacy-preserving technologies that respect data privacy at scale using cryptographic techniques.

Advisor: Nektarios G. Tsoutsos

### Department of Informatics & Telecommunications, University of Athens

Athens, Greece

MASTER OF SCIENCE (MSC) IN COMPUTER SCIENCE

October 2016 – September 2018

Advanced topics in security & architecture, operating & distributed systems, programming languages, algorithms & data structures.

### Department of Informatics & Telecommunications, University of Athens

Athens, Greece

BACHELOR OF SCIENCE (BSc) IN COMPUTER SCIENCE

October 2012 – September 2016

## Publications (Selected)

### JOURNAL ARTICLES (SELECTED)

- D. Mouris, L. Piske, P. Sarkar, N. Trieum, and Mehmet Ugurbil. “**Paras: Actively Secure Two-Server Private Histograms**” in *IACR Communications in Cryptology (IACR CiC)*, 📄 vol. 3, No. 2., 2026
- C. Gouert, D. Mouris, and N. Tsoutsos. “**HELM: Navigating Homomorphic Encryption through Gates and Lookup Tables**” in *IEEE Trans. on Information Forensics and Security (TIFS)*, 📄 vol. 20, pp. 2822–2835, 🌐 2025
- J. Gao, Y. Palanikumar, D. Mouris, D. Nguyen, and N. Trieu. “**SecurED: Secure Multiparty Edit Distance for Genomic Sequences**” in *Proc. Priv. Enhancing Technol. (PoPETs)*, 📄 vol. 2025, no. 2 2025

- **D. Mouris**, C. Patton, H. Davis, P. Sarkar, and N. Tsoutsos. “**Mastic: Private Weighted Heavy-Hitters and Attribute-Based Metrics**” in *Proc. Priv. Enhancing Technol. (PoPETs)*,  vol. 2025, no. 1,  *IETF - CFRG*,  2025
- **D. Mouris**, P. Sarkar, and N. Tsoutsos. “**PLASMA: Private, Lightweight Aggregated Statistics against Malicious Adversaries**” in *Proc. Priv. Enhancing Technol. (PoPETs)*,  vol. 2024, no. 3,  *IETF - CFRG*,  2024
- **D. Mouris**, D. Masny, N. Trieu, S. Sengupta, P. Buddhavarapu, and B. Case. “**Delegated Private Matching for Compute**” in *Proc. Priv. Enhancing Technol. (PoPETs)*,  vol. 2024, no. 2,  *Blogpost*,  2024
- C. Gouert, **D. Mouris**, and N. Tsoutsos. “**SoK: New Insights into Fully Homomorphic Encryption Libraries via Standardized Benchmarks**” in *Proc. Priv. Enhancing Technol. (PoPETs)*,  vol. 2023, no. 3, pp. 154–172, ,  2023
- **D. Mouris** and N. Tsoutsos. “**Zilch: A Framework for Deploying Transparent Zero-Knowledge Proofs**” in *IEEE Trans. on Information Forensics and Security (TIFS)*,  vol. 16, pp. 3269–3284,  2021

### CONFERENCE PAPERS (SELECTED)

- C. Gouert, K. Lee, **D. Mouris**, J.L. Watson, Y. Tselekounis. “**StOMR: Stateful Oblivious Message Retrieval**” in 2026 *ACM SIGSAC CCS*, ,  2026
- M. Ugurbil, **D. Mouris**, M.B. Santos, J. Cabrero-Holgueras, M. de Vega, and S. Sengupta. “**Fission: Distributed Privacy-Preserving Large Language Model Inference**” in 2025 *IEEE CNS*, ,  2025
- M.B. Santos, **D. Mouris**, M. Ugurbil, S. Jarecki, J. Reis, S. Sengupta, and M. de Vega. “**Curl: Private LLMs through Wavelet-Encoded Look-Up Tables**” in *CAMLIS 2025*, ,  2024

### THESES

- **D. Mouris**. “**Private And Verifiable Computation**” in *Ph.D. Thesis*, Advisor: N.G Tsoutsos,  2024
- A. Giannopoulos and **D. Mouris**. “**Privacy preserving medical data analytics using secure multi party computation. An end-to-end use case**” in *M.Sc. Thesis*, Supervisors: Y. Ioannidis and M. Garofalakis, ,  2018

### INTERNET DRAFTS (IETF)

- H. Davis, **D. Mouris**, C. Patton, N.G. Tsoutsos. “**The Mastic Verifiable Distributed Aggregation Function (VDAF)**” in *Internet Research Task Force (IRTF)*, Crypto Forum Research Group (CFRG), , ,  2024 – Now

## Technical Skills

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |                                                                                                                                                                                         |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AI &amp; Automation</b>   | Agentic Workflows  , Model Context Protocol (MCP) Integration                                                                                                                                                                                                                                                                                                                                                                                                                     |                         |                                                                                                                                                                                         |
| <b>Programming Paradigms</b> | Agentic  , Procedural  , Object Oriented  , Logic  , Functional                                                                                                                                                            |                         |                                                                                                                                                                                         |
| <b>Programming Languages</b> | Rust  , C  , C++  , Java  , Python  , Go  , Haskell  |                         |                                                                                                                                                                                         |
| <b>Parallel Programming</b>  | POSIX processes & threads  , MPI  , Open MP                                                                                                                                                                                                                                                                                                                                                      |                         |                                                                                                                                                                                         |
| <b>Assembly Languages</b>    | x86/x64  , MIPS                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Database Systems</b> | SQL  , MySQL  |
| <b>Version Control</b>       | Git  , Mercurial                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Languages</b>        | Greek ( <i>native</i> ), English ( <i>fluent</i> )                                                                                                                                      |

## Teaching Assistant

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|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>U. of Delaware</b> | Reverse Eng. & Pen. Testing (Fall '21, '22), Microprocessor Sys. (Fall '20), Applied Cryptography (Spr. '20, '21), Secure Soft. Design (Spr. '20, '21), Embedded Systems Sec. (Fall '19 – '21) |
| <b>U. of Athens</b>   | Sys. Prog. (Spr. '17), Logic Prog. (Spr. '17), Intro to Prog. (Fall '14 – '17), Operating Systems (Fall '16)                                                                                   |

## Honors & Awards

|      |                                                                                                                 |                      |
|------|-----------------------------------------------------------------------------------------------------------------|----------------------|
| 2023 | <b>Conference Stipend</b> , Travel stipend for Privacy Enhancing Technologies Symposium (PETS)                  | <i>Lausanne</i>      |
| 2021 | <b>Student Conference Grant</b> , Registration for ACM Symp. on Computer and Com. Security (CCS)                | <i>Virtual</i>       |
| 2021 | <b>1<sup>st</sup> for outstanding research presentation</b> , IEEE Reliable and Resilient Digital Manufacturing | <i>Virtual</i>       |
| 2020 | <b>Scholarship</b> , Outstanding Academic Performance Scholarship, from the Gerondelis Foundation               | <i>Grant \$5,000</i> |
| 2020 | <b>DAC Young Fellow Program</b> , 57 <sup>th</sup> Design Automation Conference, July 20-24                     | <i>Virtual</i>       |
| 2016 | <b>Scholarship</b> , Programming Languages Mentoring Workshop (PLMW) at SPLASH 2016, 10/30-11/4                 | <i>Amsterdam</i>     |

## Community Service

- **Global Co-lead** in BlueHens Capture The Flag (CTF) Competition - UDCTF. 2021, 2022
- **Global Co-lead** in CSAW Cybersecurity Games & Conference – Embedded Security Challenge 2020 – 2023, 2025

## Work Authorization

US Permanent Resident via National Interest Waiver based on merit.